

Detecting and Analyzing Limit Cycles in Agent-based Models: the Case of the El Farol Bar Problem

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Abstract

This study introduces a novel data-driven methodology to detect and classify limit cycles in agent-based simulations. The methodology is tested on an epidemiological expansion of the classic El Farol Bar problem, where agents not only decide whether to attend a crowded event based on bounded rationality and past attendance, but also face the risk of infection; hence, a cyclic behavior could emerge in the two-dimension space. A computational experiment was conducted, exploring the parameter space to assess the existence and the causes of these cyclic behaviors. Limit cycles were identified via autocorrelation-based seasonality analysis of two key time series — attendance and infection levels — without relying on analytical solutions of the underlying dynamics that were not feasible. The method enabled the classification of recurring patterns in phase space, revealing the emergence of 23 distinct cyclic shapes. These patterns were then analyzed in relation to the system parameters that generates them, allowing for the construction of a proximity network of behaviors. The results demonstrate how even simple agent-based models can produce a rich taxonomy of cyclic phenomena, with potential implications for both system dynamics and public health policy; also, they pave the way for more extensive applications of cycle detection in models where analytical tools are inapplicable.

1 Introduction

Simulation models are generally used for replicating complex real-world phenomena, providing insights and supporting informed decision-making [1, 2]. These models enable researchers and decision-makers to experiment virtually with different scenarios [3], assessing potential outcomes without the constraints and risks associated with real-world trials [?]. In healthcare management, simulation models have been employed effectively to manage hospital flow in specific regions [4] or to predict pharmaceutical consumption [5]. Agent-based models are employed in urban planning and transportation to facilitate the analysis of citizens' behaviors, taking into account individual actions and interactions [6], while system dynamics models have proven valuable for examining market behaviors and economic cycles [7], incorporating feedback loops, delays, and nonlinear relationships [8].

However, it is crucial to recognize that computational simulation models are not inherently valuable or useful on their own [1]. Their real utility depends on how

effectively their behavior, particularly the outputs they generate, is studied and interpreted [9,10]. A model, no matter how sophisticated or well-calibrated to a real-world scenario [11], can only offer meaningful support if the results it produces are critically analyzed and understood in context [12]. Therefore, rigorous analysis, careful output selection, and thoughtful interpretation are essential to fully harness the potential of simulation models and to translate their computational insights into practical, real-world value [13]. Several techniques have been specifically developed to explore the behavior of computational models systematically and enhance the reliability of their results, such as sensitivity analysis [9] and Latin Hypercube Sampling [14].

Nevertheless, in computational simulations, emergent phenomena often arise from complex interactions that are not easily predictable or reducible to closed-form expressions [15], and the understanding of behaviors like limit cycles typically relies on empirical exploration [16].

Limit cycles are defined as closed, periodic trajectories in a system's phase space towards which nearby trajectories converge or diverge [17]. These cycles are defined by their capacity for self-sustained oscillations, characterized by a constant amplitude and periodicity that remains unaltered, irrespective of the initial conditions from which they are initiated [18,19]. The concept of limit cycles was first formally established by Henri Poincaré in the late 19th century [20], and later gained broader attention through David Hilbert's list of 23 mathematical problems presented at the Second International Congress of Mathematicians [21]. They have since then been identified in a wide range of scientific domains, including physics, engineering, biology, neuroscience, economics, and epidemiology [17,22–27]. No specific technique or set of techniques currently exists for fully analyzing the emergence of limit cycle behaviors in non-analytical simulation models [28,29].

This paper faces the issue leveraging a pre-existing model [30] that introduces an epidemiological extension to the classic El Farol Bar problem [31], where individuals not only decide whether to attend the venue based on individual preferences and crowd dynamics but also face the possibility of becoming infected or transmitting the infection to others [32]. This El Farol bar problem extension was implemented into an agent-based model (ABM) [33,34], and the data generated by a model's behavior exploration was collected and processed to identify potential limit cycles using a custom data-driven methodology, extracting patterns directly from the simulation data.

The paper presents three main contributions. First, it introduces a novel methodology for the detection of limit cycles. The methodology assumes that the presence of a limit cycle requires cyclic seasonal interactions among at least two variables, and that this seasonality can be identified empirically via autocorrelograms, by locating the lag at which the autocorrelation attains its maximum absolute value. Second, it proposes a novel way to analyze the effect of parameters on the resulting limit cycles shapes emerged, leveraging on the rich dynamics of the underlying model. It provides initial insights into how such shapes may interact, overlap, or influence one another. Finally, it discusses the practical implications of the results, demonstrating how the analysis of limit cycles can yield actionable information for practitioners, domain experts, and policy makers, using a case study. While the case study naturally suggests potential implications for policy makers, the primary contribution of this work is methodological.

The paper is structured as follows. First, the methodology is detailed, including an overview of the agent-based model, the data generation process, and the custom method

developed to identify limit cycles. Next, the results are then presented, followed by a conclusion that discusses and synthesizes the findings and their implications.

2 Background

2.1 Simulation and limit cycles

Simulation techniques are one of the standard methodologies for analyzing and managing complex systems characterized by non-linear interactions, feedback loops, and emergent behaviors [35]. ABMs are particularly effective in capturing the heterogeneity and autonomy of individual actors within systems [36]. In healthcare, ABMs have been successfully applied to optimize emergency department operations and improve decision-making for patient flow and staff allocation [18,37]. In transportation, agent-based urban modeling has been used to incorporate demographic and spatial data into mobility simulations, offering insights for planning and infrastructure development [38,39]. In economics, ABMs simulate market behaviors and policy effects, supporting decisions in urban growth and macroeconomic planning [40]. Also, hybrid approaches integrating ABM and System Dynamics offer a powerful toolkit for understanding multi-scale dynamics, as demonstrated in studies of organizational learning and policy evaluation [41].

Parameter exploration could lead to the discovery of peculiar behavior [42], such as limit cycles [17,43]. In analytically tractable dynamical systems, the identification of limit cycles is facilitated by a suite of classical and modern techniques, each adapted to the system's structure and dimensionality. The foundational Poincaré-Bendixson theorem allows detection in two-dimensional systems by analyzing the phase space and ruling out other long-term behaviors [44]. Tools such as Dulac's criterion and Liénard systems offer analytic routes to proving the existence or absence of cycles in polynomial systems [45]. Perturbation techniques and harmonic balance methods yield approximations for the shape and stability of limit cycles, especially in weakly nonlinear systems [45,46]. Additionally, invariant curve constructions and Lyapunov function-based methods enhance detection and stability assessment [47,48].

Identifying limit cycles in non-analytical simulation models such as ABMs poses significant challenges due to the absence of closed-form equations and the inherent stochasticity and discrete nature of these systems [16,49,50]. So, classical analytical methods are largely inapplicable to models driven by local interactions and emergent behavior [43,51]. Moreover, ABMs often exhibit transient dynamics and noise-driven variability, making it difficult to distinguish true periodic behavior from pseudo-cycles or artifacts of finite simulation time [52,53]. Possible approaches include geometric visualization combined with computer graphics for inferring cyclic behavior in high-dimensional state spaces [29], as well as harmonic linearization and numerical continuation techniques [54]. Also, neural modeling has explored limit cycle storage as a form of dynamic memory [55].

2.2 El Farol Bar problem

Since its formulation by Arthur [31], the El Farol bar problem has inspired a broad spectrum of modeling approaches aimed at understanding coordination under bounded rationality and limited resources. Central to these variants is the specification of the agents' payoff function, which determines how the problem is interpreted. One major class conceptualizes the bar as a scarce resource to be distributed fairly among agents,

leading to formulations where agents aim to secure access while minimizing waste [56, 57]. A second interpretation aligns the El Farol model with minority games, in which agents receive higher payoffs when acting contrary to the majority [58]. A third perspective treats the problem as one of infrastructure congestion, wherein any excess attendance yields a universally negative payoff [59, 60]. This interpretation emphasizes the discontinuity of payoffs around the threshold and the systemic need for differentiated expectations to avoid coordination failure [31].

Parallel to variations in payoff structures, El Farol models differ significantly in how agents make decisions [61]. Arthur’s original predictor-based method involves agents selecting heuristics that forecast bar attendance from past rounds, and updating them based on predictive accuracy [31, 62]. This “best predictor” approach has been widely adopted [57, 63] and extended with variations on predictor memory and selection rules [64]. Other researchers have proposed fixed cyclic attendance strategies [59], autoregressive predictors evolved via genetic algorithms [65], and social-network-based coordination with fairness-aware agents [66]. Still others have examined probabilistic decision-making, either via static attendance probabilities or reinforcement learning schemes that adapt based on past outcomes [60, 67].

The El Farol Bar problem has inspired a wide array of extensions that explore how coordination emerges in complex systems under limited information and bounded rationality. Strategic group formation has been introduced, allowing agents to pool strategies and make collective decisions, though this can backfire due to the compounded impact on congestion [68, 69]. Extensions into reinforcement learning and meta-strategy evolution examine how agents adapt over time to reduce attendance fluctuations [70, 71]. The problem has also been framed as an iterated N-person game, offering insights into self-referential expectation dynamics and analogies with financial markets [72]. Network-based approaches represent another extension, where social structure and preference heterogeneity drive coordination outcomes [?]. Cognitive extensions include emotion-aware agents and belief-driven behavior, reflecting decision-making in more human-like decentralized markets [73]. Lastly, the model has been generalized to new settings such as the Kolkata Restaurant Problem, which expands the scope from a single venue to many competing resources [74], and to internet congestion models [75].

3 Methods

The methodology section is organized into three parts. The first part provides a brief description of the model selected as the case study, with reproducibility ensured by referring to previous work. The second part details the experimental configuration employed to generate data from the model. Finally, the third part presents the specific technique developed in this study for the detection of limit cycles from the generated data. The model, the data and the results are all available at the following repository: <https://anonymous.4open.science/r/elfarolllimitcycles-7E5A/>

3.1 Agent-based model description

The agent-based model incorporates an epidemiological extension of the El Farol Bar problem [31], where agents not only decide whether to attend the weekly event but can also become infected if they participate and encounter infected individuals. For clarity, the description is organized into two sections: the social dynamics of the system and the epidemiological dynamics. For sake of brevity and to focus on the actual results of this work, this paper provides only a brief overview of the model; for a more detailed and

reproducible description, refer to the previous work [30].

The choice of relying on this pre-existing model is motivated by reasons of scientific efficiency and reproducibility: it allows us to focus on the methodological contribution without redefining a new system from scratch, while ensuring that the underlying dynamics are already documented and replicable. Moreover, the model is particularly well suited for the present purpose, as it generates only two output variables and in its non-epidemiological formulation it is already known to exhibit cyclic behaviors, providing an ideal testbed for limit cycles detection.

3.1.1 El Farol bar model

The proposed model features a single type of agent, standing for individuals who may choose to attend the bar during any given time step. The agents' behavior follows the assumptions of the original El Farol Bar problem. First, each agent's only decision each week is whether to attend the bar and this decision is always carried out. Second, agents interact with one another exclusively at the bar, meaning after they have already made their attendance decision. Third, agents prefer to attend the bar when it is not overly crowded and seek to do so as often as possible; therefore, each week, an agent decides whether to attend based on its expectation of the total number of attendees.

Information about past attendance is crucial in forming agents' expectations, as it is used to estimate the number of agents likely to attend the bar in the following week, according to the following mechanism. For agents who attended the event, the new memory value corresponds to the number of agents present at the bar at the same time step. For agents who did not attend, the new memory element is a random value sampled from a uniform discrete distribution between 0 and $n - 1$, representing an educated guess made by agents who cannot communicate with one another.

Each agent i (with i ranging from 1 to n , the total number of agents) forms an expectation about how many agents will attend the bar at the next time step by memorizing the number of agents present during the last x times it attended, where x denotes the memory length, and by assigning weights to generate a prediction, where the weight assigned to the last observation is m . The agent's decision to attend or not is made by comparing an attendance threshold with the expectation of future attendance. The attendance threshold t_a is a model parameter that represents the venue saturation level beyond which agents would find being at the bar unpleasant and therefore choose not to attend.

3.1.2 Epidemiological extension

In epidemiological models, agents are typically, although not necessarily, classified into one of three states: susceptible, infected, or recovered. These types of models capture the possibility of acquiring immunity following an infection and eventually becoming susceptible again, thus representing diseases in which immunity, whether naturally acquired or vaccine-induced, can be gained and subsequently wanes over time.

The epidemiological part of the model assumes that individuals can be in one of these states, and that disease transmission can occur only during attendance at the event, with a contagiousity α . Furthermore, the epidemic sub-model is based on several modeling hypotheses. First, the model assumes that the disease is non-lethal; agents cannot die from the infection, thus the total number of individuals in the system remains constant. Second, it is assumed that there are no long-term physical or

psychological effects following infection, so the recovered agents retain their full ability to engage in normal activities, such as attending the bar, indicating that the disease does not result in lasting physical or psychological impairments. Third, the only epidemiological variable an agent considers when deciding whether to attend the bar is their own contagion level c_i : if it is too high, they may choose not to attend due to the symptoms. Consequently, a high infection level can reduce attendance without involving mortality or permanent effects, at least in the short term. Fourth, agents are unaware of the global epidemic's existence and dynamics, having no information about the number of infected individuals and not factoring infection risk into their decision-making process. They are aware only about their individual status. This results in a lack of adaptive behavior in response to the epidemic, which significantly affects the model's predictions about disease spread. Finally, the contagion process is assumed to be uniform across all agents, featuring a constant duration and an initially uniform level of infectiousness.

Whenever an agent becomes infected, the infection follows with a linear dynamic, and the agent undergoes a recovery process lasting d_i time steps during which c_i decreases by $\frac{1}{d_i}$ at each time step. Two thresholds, t_s and t_c , play a crucial role in shaping agent behavior and the dynamics of infection spread. The first threshold t_s stands for the infection level at which an agent develops symptoms severe enough to discourage attendance at the bar; on the other side, t_c is the infection level above which agents are capable of transmitting the disease. The infection is primarily spread by agents whose contagion level falls between these two thresholds, that is, when $t_c < c_i < t_s$. After recovery, agents acquire temporary immunity lasting t_r time steps.

3.2 Experimental setup

The data are obtained by running the agent-based model 100,000 times to gather outputs across different combinations of parameters [14]. The specific methodology applied is random grid sampling, where results are collected from experiments generated using parameters drawn from a finite parameter space. The rationale behind employing this technique is to facilitate the exploration of system behavior across various parameter combinations, particularly in cases where no specific behavior is expected or targeted. Table 1 lists the parameters tested in the simulation along with their explored ranges. Each parameter is sampled from a uniform distribution.

Table 1. Description of the model parameters and their sampling ranges used in the grid exploration of system behavior

Parameter	Description	Sampling range
t_a	Threshold of the expected share of agents beyond which an agent decides not to attend the event because the bar would be too crowded	[0.01, 1.00], step 0.01
d_i	Time duration of the infection process	[1, 10], step 1
t_s	Threshold of infection above which infected agents develop symptoms and refrain from attending the bar	[0.01, 1.00], step 0.01
α	Contagiousness level of the considered disease	[0.01, 1.00], step 0.01
t_c	Threshold of infection below which an agent is no longer able to transmit the disease	[0.01, 1.00], step 0.01
m	Weight assigned to the most recent memory in the agents' decision-making process	[0.01, 1.00], step 0.01
t_r	Duration of the recovered status	[1, 10], step 1
n	Total number of agents in the simulation	[2000, 2000]

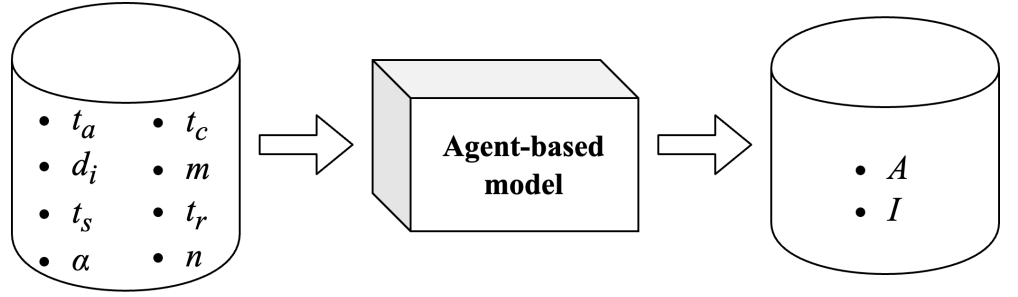


Fig 1. Black box representation of the experimental setup

The experiment produces two distinct time series: the time series I , representing the number of infected agents at each time step, and the time series A , representing the number of agents attending the event at each time step. Figure 1 depicts the experiment.

3.3 Limit cycle detection

As mentioned in the introduction, the aim of this paper was to identify interesting patterns in the relationship between infections and participation in social events. From the perspective of dynamical systems, one intriguing aspect would obviously have been the inclusion of a set of agents engaging in various activities, and how the dynamics of this activity varies.

A new methodology was then developed to identify the limit cycles present in the obtained simulations. This approach was, of course, "data-driven" rather than "equation-driven", as the model is not defined at the macro level, making it impossible to derive the Lyapunov exponents of the underlying dynamical system. Thus, an alternative method needed to be identified. A recently developed methodology demonstrates how to determine the spaces, or at least some of the spaces, where limit cycles appear through specific experiments [16]. However, this approach was unsuitable for our case for two reasons. First, we did not aim to conduct an ad hoc experiment to identify limit cycles but instead sought to broadly explore the parameter space and then observe what occurred in individual simulations with sampled parameters. Second, this methodology is computationally very demanding.

The following methodology was therefore adopted. It begins with the assumption that a limit cycle presupposes cyclicity between at least two variables. This implies that the n curves under consideration must exhibit seasonality and, likely, share the same period of seasonality. Additionally, it is known that the seasonality of a time series can be identified from the available data using an autocorrelogram, by determining the period at which the autocorrelation is maximized in absolute value.

Here, the approach followed was precisely as described. Simulations were conducted, collecting the values of the curves of the two global system variables: the total number of agents visiting bar A and the total number of infected agents I . Subsequently, the autocorrelograms of the two curves, $ACF[A]$ and $ACF[I]$, were constructed. These autocorrelograms were analyzed for two criteria: first, both curves needed to have a maximum autocorrelation value greater than 0.7, as a strong autocorrelation was desired; second, the periods (the lag time) of the two curves had to be approximately equal, allowing for a margin of error of one time step. This margin was included to

account for noise in the curves, which might otherwise obscure interesting behaviors. The curves identified through this process were classified as exhibiting limit cycles, while the others were not.

This methodology naturally has its limitations, chief among them being that the conditions it identifies are likely sufficient but certainly not necessary. A curve exhibiting these characteristics will display a behavior in state space that resembles a limit cycle, but this does not exclude the possibility of other limit cycles that go undetected. The goal of this methodology is not to identify every single limit cycle but to capture the majority of them. Further analyses should be conducted to assess its accuracy in different contexts.

4 Results

4.1 Limit cycles emergence

As mentioned in the methodological section, the autocorrelation of A and I were used to assess whenever a possible limit cycle emerges in the system behavior. The values of $ACF[A]$ and $ACF[I]$ are presented in Figure 2, where it is observable how a noticeable proportion of the simulations result in early termination of the infection, with no persistent cyclic behavior. These dots are clustered in areas of low or negative autocorrelation, particularly for A , suggesting that irregular or aperiodic dynamics are more prone to leading to infection extinction. This highlights the system's sensitivity to non-cyclic patterns, where chaotic or low-autocorrelation behaviors fail to sustain long-term infection. Also, the presence of red points, where A and I do not share nearly identical periods, suggest that also non-synchronized dynamic between the two variables exist. This may indicate a deeper structural property of the system, wherein the fluctuations in agent behavior (A) and infection's spread (I) can be independent and not necessarily co-evolve.

Beyond the impressions, it is noteworthy to assess the numerosity of each combination. A matrix is then defined and shown in Figure 3, counting the simulations in which the infection either survives or does not, as well as the presence or absence of cyclic behavior. Each labeling number stands for the number of simulations with that specific combination. The majority of simulations result in the infection's demise without exhibiting cyclic dynamics, while of the 4,160 simulations in which at least one infected agent was present at the end of the 200 time steps, approximately 24.3% also exhibited cyclicity (corresponding with the light blue dots in Figure 2 and the upper right section of Figure 3). Noticeably, a 0.032% of the simulation results demonstrated both cyclic behavior with infection extinction and a co-effect of epidemic and social features.

The simulation outcomes can be classified into two distinct groups of primary interest for the scope of this paper, based on their characteristics and behaviors:

1. Simulations S_a : all the simulation results, serving as a comprehensive dataset to which to make comparisons.
2. Simulations S_c : simulations where $\max(ACF[A]) > 0.7$ and $\max(ACF[I]) > 0.7$, and $\arg\max(ACF[I]) - 1 \leq \arg\max(ACF[A]) \leq \arg\max(ACF[I]) + 1$.

We focus on S_c because the purpose of this research is to discover, analyze and interpret the cyclic pattern between I and A .

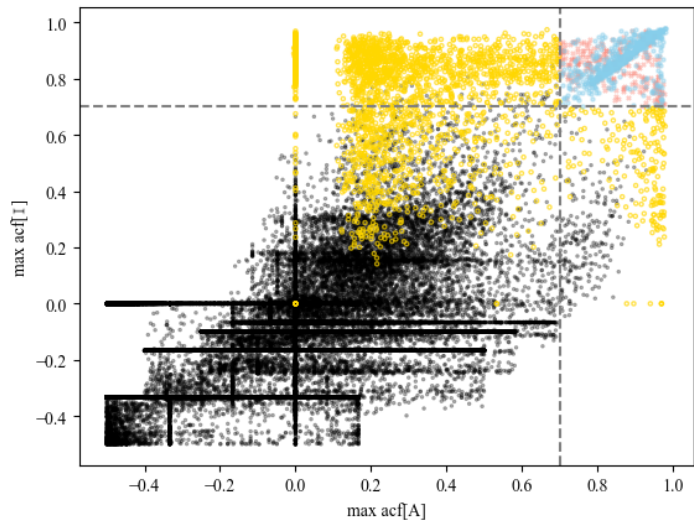


Fig 2. The scatter plot depicts the autocorrelations of the time series of I and A , with each point representing a distinct simulation. The light blue and red points represent runs where both the $\max(ACF[A])$ and $\max(ACF[I])$ exceed 0.7. However, in the runs represented by light blue points, the period of A and I (which is the lag with the highest autocorrelation) is identical, with a possible error of 1. The yellow dots represent simulations in which no limit cycle was identified, and the infection persisted until the conclusion of the simulation. Conversely, simulations without limit cycles and that terminated before reaching the end are indicated by black dots.

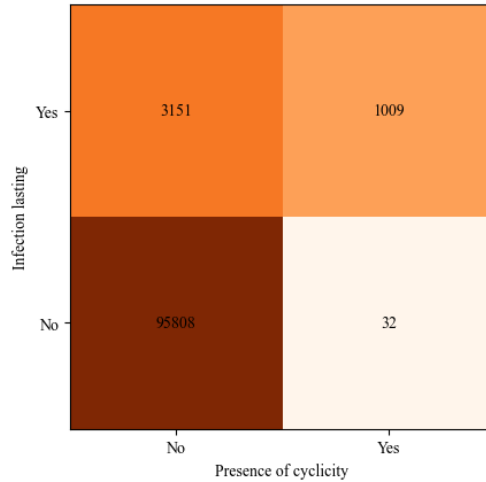


Fig 3. The simulations are distributed into four possible cases: infection lasting until the final time step on the y-axis, and the presence of cyclicity (as defined in the Methodology section) on the x-axis. The colours highlight an increased or decreased number of occurrences.

par	$E[par S_a]$	$E[par S_c]$	$\frac{E[par S_c]-E[par S_a]}{E[par S_a]}$
t_a	0.50	0.61	0.22
t_s	0.50	0.14	-0.72
d_i	5.51	7.59	0.38
m	0.50	0.44	-0.12
t_c	0.50	0.31	-0.38
t_r	5.01	4.28	-0.15
α	0.26	0.30	0.17
n	2252.35	2655.10	0.18

Table 2. Parameters' effects comparisons across the two identified scenarios.

Table 2 presents a comparative analysis of various parameters across these two simulation scenarios. The threshold t_a is higher in S_c than in S_a (a 22% increase), indicating that simulations where agents avoid attending the event after reaching a higher threshold of expected agents are more likely to produce a cyclic behavior, with a stronger effect in cyclical cases, suggesting that the presence of a more "picky" behavior creates conditions conducive to sustained infection spread and, especially, cyclicity. Also, t_s is much lower in S_c , indicating that a scenario with cyclical behavior appears more likely when the threshold of perception of infection is low. This makes sense, given that to be in S_c the infection has to last, and this happens more likely when agents attend the event even if infected. The same explanation relates to d_i and n .

4.2 Limit cycles behavior

To observe the system's behavior, an initial approach involving a tripartite graphical representation was adopted, as illustrated in Figure 4, which displays the behavior in three distinct ways. First, the time series for both A and I were plotted (Subfigure a) over the entire duration of the simulation. Subsequently, A and I were represented on the state space, both with the full dataset and with only the second half of the data, i.e., the time steps following the first 100 time steps (subfigures b and c, respectively). This approach was selected to identify the existence of non-point equilibria while either mitigating or disregarding the influence of initial conditions. As can be observed, the full dataset graph exhibits an ill-defined shape, whereas the truncated version (on the left) is much clearer. Its analysis allows for the conclusion that cyclical behavior is present.

Figure 5 displays a series of heatmaps, each representing the relationship between one of the model's key parameters and the seasonality of the detected limit cycles, showing regions of the parameter-seasonality space where cycles of certain lengths are more likely to occur. Looking at the panels, it is possible to say that the seasonality of the cycles is not uniform with respect to the parameters. Also, one could argue that the cyclical structure of the system does not depend on a single parameter, but is rather the result of nonlinear interaction.

For example, in panel (d), as contagiousness α increases, longer cycles (higher values of $\arg \max(\text{ACF}[I])$) become more frequent, but only until a certain level, since there are just few simulation in S_c with a value of $\alpha > 0.45$. Similarly, panel (h) shows that the number of agents increases that mean length of cycles. Also, a low value of the infection perception threshold is associated with shorter and more regular cycles, while higher values lead to more varied or less cyclical behaviors. This type of analysis makes it possible to connect the emergent dynamics to specific parameter configurations, offering both methodological and practical insights for the management of social and

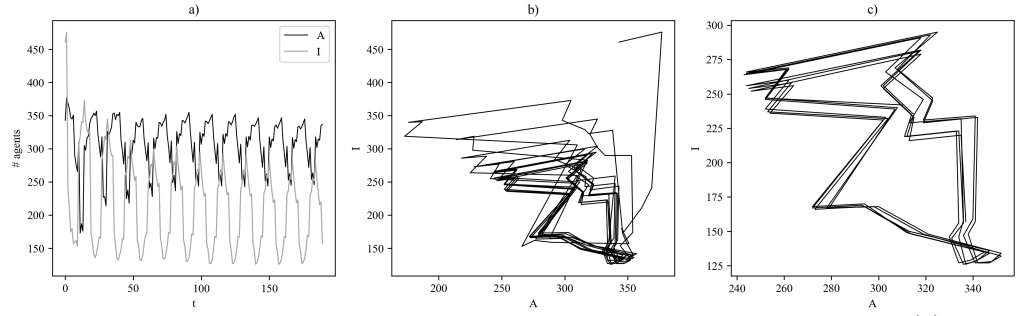


Fig 4. An example of the dynamics of attendance A and infections I . Panel (a) displays the time series of the two variables over the entire simulation, showing the co-fluctuations between the number of attending agents and infected agents. Panel (b) depicts the system trajectory in the A – I state space across all time steps. Panel (c) reports the same state-space trajectory after discarding the first 20% of the time steps, in order to remove transient dynamics and highlight possible cyclic patterns with reduced noise.

epidemic systems. Finally, in panel (b) and (g) the time-related parameters (d_i and t_r , respectively infection and immunity duration) have a diagonal relationship with seasonality, given that not only the average period is higher when the parameters are higher, but also it looks that there is a linear positive relationship, at least above a certain values. Again, this makes sense: the more it takes to each individual agents to recover or to be susceptible again, the longer the overall system would take to come back to the same status.

A preliminary analysis revealed the presence of several recurring but distinct patterns. Consequently, these patterns were manually categorized into groups. At the end of this process, 23 distinct shapes were identified, accounting for 64.19% of the simulations in scenario S_c . The remaining cases included patterns that were either poorly defined or displayed intermediate forms between two or more clusters, and for this reason, they were excluded from the analysis. Nevertheless, all figures, along with their classifications, are available in the online repository. Figure 6 illustrates the distribution of the 25 identified categories (the 23 clusters, the mixed cases, and the undefined patterns), while Figure 8 presents, among other elements, representative simulations of the identified forms.

The presence of such diverse patterns within the same system is, in itself, intriguing and not obvious. The model’s scheduling is not excessively complex and the developed model is relatively simple. Nevertheless, the complexity of interactions permits the emergence of a wide range of highly interesting and diverse behaviors. After confirming that this significant variability was not influenced by specific initial conditions or the random seed used to generate the data (a conclusion supported by a preliminary analysis), it was possible to take a further step and investigate the underlying causes. Specifically, the focus shifted to identifying which combinations of environmental parameters led to the emergence of specific cyclic behaviors. To this end, the mean values of the parameters for each simulation cluster were calculated, as shown in Table 3, alongside the average seasonality of the cycles $\argmax(AFC[C])$.

Certain patterns exhibit parameter values of particular interest. For instance, the average threshold of attendance t_a is exceptionally high for the “flag” shape. We interpret this to mean that a very high participation threshold can lead to a fuller

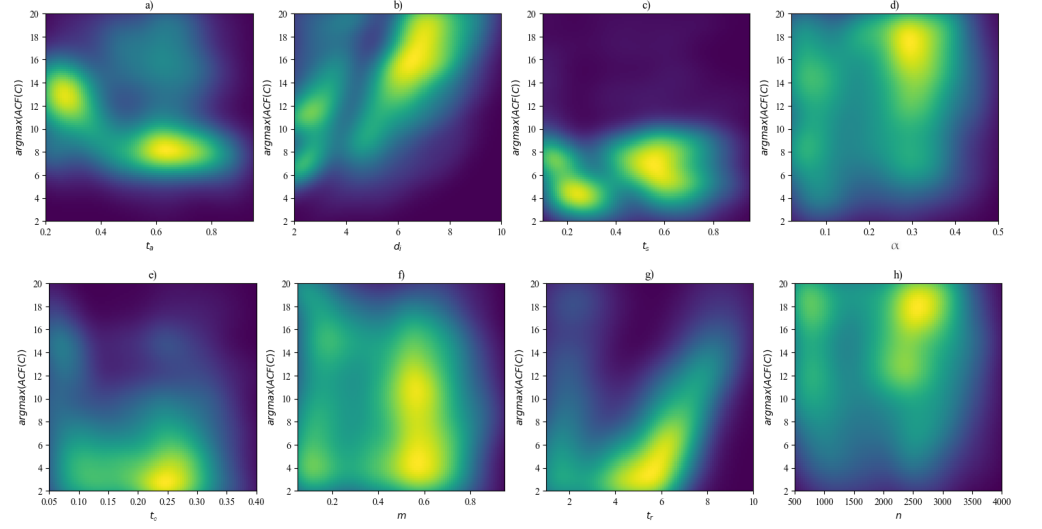


Fig 5. relationship between the seasonality of the limit cycle and the value of each parameter, where color intensity indicates the number of simulations in that area of the parameter space (lighter areas correspond to a higher number of simulations; darker areas to a lower number).

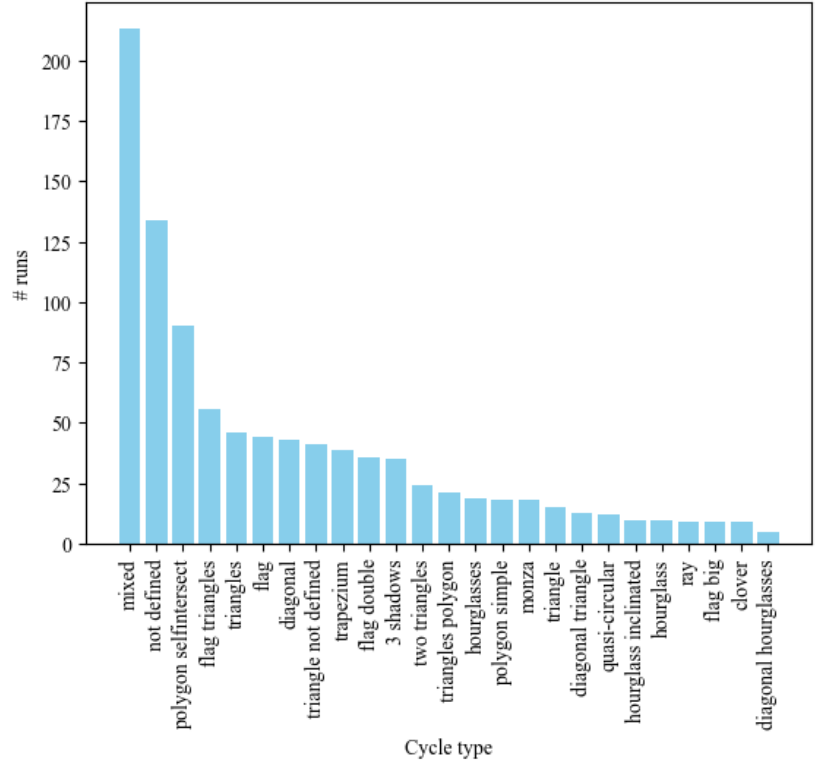


Fig 6. Distribution of the different limit cycles shapes in S_c

	t_a	t_c	d_i	m	t_s	t_r	α	n	Z
polygon self-intersect	0,53	0,11	7,93	0,43	0,31	3,17	0,31	2650	12,08
flag big	0,56	0,16	7	0,68	0,42	5,44	0,34	2611	13,44
flag	0,91	0,08	8,25	0,43	0,22	3,02	0,28	2500	13,14
diagonal hourglass	0,55	0,27	5,6	0,10	0,39	4,2	0,34	2700	3,6
quasi-circular	0,73	0,11	7,83	0,43	0,37	3,92	0,24	3333	11,67
polygon simple	0,55	0,12	7,56	0,49	0,35	4,28	0,34	2388	12,44
hourglass	0,60	0,16	6,3	0,38	0,26	5,1	0,27	2850	7,7
clover	0,69	0,08	8,67	0,61	0,26	2,89	0,27	2888	12,56
triangles	0,55	0,14	6,63	0,47	0,23	5,26	0,34	2532	11,8
hourglass inclined	0,74	0,10	7,4	0,65	0,17	5,3	0,30	2650	9,38
triangle	0,60	0,13	5,4	0,37	0,24	2,13	0,24	2866	5,6
trapezium	0,60	0,09	6,82	0,54	0,18	4,77	0,28	2666	3,6
flag triangles	0,78	0,09	8,27	0,46	0,20	1,91	0,27	2232	5,08
monza	0,48	0,18	7,78	0,32	0,39	7,06	0,36	2722	11
flag double	0,48	0,14	7,61	0,39	0,41	4,31	0,28	3055	15,83
triangle not defined	0,66	0,18	6,59	0,58	0,27	5,46	0,28	2731	12,92
3 shadows	0,46	0,21	7,06	0,39	0,71	4,49	0,32	3328	3,88
triangles polygon	0,52	0,16	7,57	0,37	0,27	4,67	0,31	2190	12,54
hourglasses	0,47	0,25	5,84	0,44	0,35	4,16	0,38	2631	11,95
two triangles	0,50	0,11	8,21	0,50	0,19	2,92	0,34	2770	11,89
diagonal	0,65	0,25	5,49	0,38	0,46	4,42	0,33	2802	6,38
ray	0,59	0,20	5,78	0,34	0,34	1,44	0,37	2777	2,33
diagonal trapezium	0,62	0,24	7,38	0,56	0,32	3,23	0,34	2307	9,35

Table 3. Parameter values for the different forms identified. Z is $\arg \max(AFC[C])$, used exclusively for the sake of table graphical representation.

venue, resulting in highly fragmented behavior. Additionally, when a large number of people consistently visit the venue, there are periods when the entire population becomes infected, producing the linear shape on the right side of the cycle. Notably, "flag" also shows an above-average d_i value, which is similar to the "flag double" category. In "flag double" there are always periods when the venue is full, but the intermediate periods are more frequent and complex. Cycles labeled "flag big" on the other hand, have above-average values of d_i and t_a , though not as high as those of "flag". However, they also exhibit a high t_r value, which functionally causes them to behave similarly, albeit at a slower pace.

It is also observed that the recovery time t_r generally has a progressive effect on sharpening the angles of the patterns. As t_r decreases, it first creates triangular shapes, and at its minimum, it results in the "ray" pattern, which is nearly collapsed. For simulations labeled as "monza" (named after the racetrack that the shape resembles), t_r is slightly higher than in the three-flag categories, though it still maintains a phase where the entire population is infected. Overall, the emergence of such patterns is highly intriguing from a managerial perspective and, with further empirical analysis, could support policymakers. This behavior could potentially be managed by imposing a cap on the population allowed to attend social events — a feature inherently present in the model, as the number of agents is always fixed at n .

The infectivity threshold t_p , beyond which individuals refrain from attending events, is notably higher — approximately double — for the "three shadows" category, named for its resemblance to Auguste Rodin's sculpture of the same name, compared to other identified cases. In this scenario, agents visit the bar even at higher levels of contagion. As a result, the variability of A is very low, as attendance is primarily influenced by t_a , while infections progress in waves. This represents a scenario of an endemic disease that has minimal, if any, impact on individual behavior.

The two values which in theory are the most important, the contagion rate c and the number of agents n , have a limited effect on the specific shapes observed. Notably, only n appears to act as an enabling factor, though not a sufficient one, for the emergence of an almost circular shape. Such a shape indicates that changes are progressive rather than instantaneous, resembling more closely a time-based simulation rather than the event-based nature of this model.

It can be observed that for more triangular shapes, the infection duration d_i is generally lower, while in some of these cases, the contagion threshold t_i beyond which individuals become infectious is relatively high. A shorter infection duration limits the period during which transmission can occur, potentially resulting in "short" cycles that span a limited number of points. This is also reflected in the value of $\argmax(AFC[C])$, which is notably lower in cases where the polygon contains fewer points, such as triangles. However, exceptions exist, such as diagonal shapes, which suggest that what appears as two separate entities might more likely represent collapsed, more complex forms, condensed into a thicker or thinner segment. For example, the difference between shapes like "clover," "Monza," and "triangle" lies not only in their forms but also in the number of points defining the cycle. Based on this observation, an analysis was also conducted to explore potential relationships between parameter values and identified seasonality. No strong relationships were found, except for d_i and t_r . These two parameters exhibit the only cases where the absolute Spearman correlation — chosen to capture co-variations rather than strictly linear relationships — exceeds a threshold set at 0.2, with values of 41.52% and 23.64%, respectively.

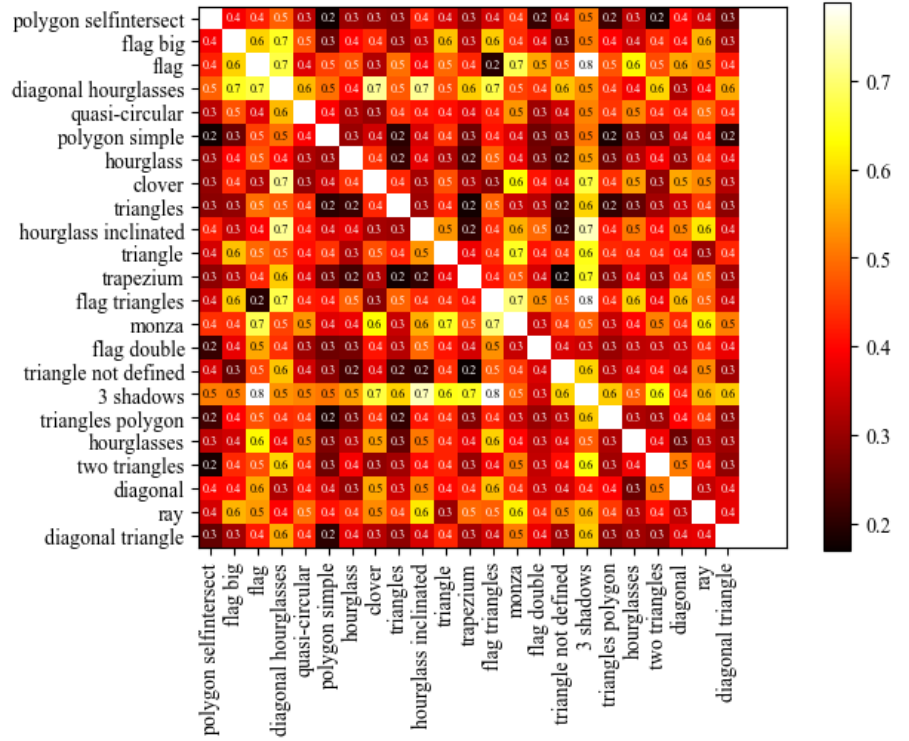


Fig 7. Heatmap of the Euclidean distances between the means of the normalized parameters for each cycle shape.

Once the discussion on how parameters give rise to different forms is complete, it becomes possible to question the relationships between these forms in terms of the average position of their generating values within the parameter space. To achieve this, an appropriate method for defining their distance must be identified. The following approach was chosen: starting from the results in Table 3, all eight parameters of interest were normalized using the minimum and maximum values observed in the experiments. Subsequently, the Euclidean distance was calculated in the resulting 8-dimensional space, where each dimension represented the average parameter value for a specific form. This metric was then used to define the proximity between observed behaviors of the limit cycles. The results can be observed in Figure 7.

For instance, it can be observed that "trapezium" is very similar to "hourglass inclined," "triangle," and "triangle not defined," three shapes to which it indeed bears a strong resemblance. At the same time, it is vastly distant from "three shadows." Thus, much like semantic similarity in the case of Large Language Models, topological proximity also conveys meaningful information.

It is possible, for example, to transition from "diagonal hourglass" to "diagonal" primarily by modifying only the social component of the system — specifically, by increasing the weight of short-term memory in the decision - making process and the threshold of venue occupancy considered satisfactory. This adjustment simplifies individual agents' decision-making behavior while preserving the diagonal shape (which reflects the interaction between the epidemiological and social components). Consequently, the entire system transitions between only two points instead of four,

effectively simplifying the overall behavior.

Similarly, it is possible to transition from "inclined hourglass" to "trapezium" primarily by modifying the crowd tolerance threshold and the weight of short-term memory — again, social components — while also slightly reducing the infection duration and the time agents remain uninfected. The key difference lies in the width of the horizontal sections of the cycle, which correspond to periods where attendance is unaffected by the epidemiological component. Although the epidemiological state fluctuates, it does not significantly influence attendance during these intervals. By lowering t_a and m , agents become less inclined to visit a venue if it was heavily crowded during previous periods. This results in a sharper reduction in attendance during periods of high infection. At the same time, a single period with a high number of infected individuals does not immediately lead to a drop in participation, creating a more gradual response in the system.

A third transformation worth reporting and briefly discussing, as it differs from the two previously examined, is the transition from "polygon simple" to "polygon self-intersect". In this case, the resulting shape in the state space begins to self-intersect when the number of time steps during which agents remain non-infectable after an infection decreases. This is accompanied, to a lesser extent, by an increase in population size, contagion rate, and a reduction in the role of short-term memory. As shown in Table 3, the number of points touched by the two cycles remains almost the same (approximately 12), but their arrangement changes. Specifically, a reduction in the refractory immunity period lengthens the transitions between states, transforming a hollow figure into a filled one. A similar effect is observed when adding agents to the system, which increases the potential jump, particularly in relation to a threshold unaffected by population size. Conversely, we propose that an increase in contagion rate makes the simple polygon self-intersect by allowing the system to reach states with intermediate values of A and I . This creates overlapping trajectories, further increasing the complexity of the cycle structure.

Finally, using this proximity metric, it is possible to construct a network of relationships among the various cycles, providing an illustrative way to visualize transitions between different forms. The following logic was adopted to construct the network: all pairs of forms were sorted in ascending order according to their distance. In each iteration, a pair was added to the network only if no existing path already connected the two nodes. If one of the nodes in the selected pair was not yet present, it was added to the network first, and the link was created afterward. The network construction concluded when all nodes were connected, meaning the giant component encompassed the entire network. This technique is not perfect, as it captures only a subset of the proximity relationships and not all possible connections. For instance, "inclined hourglass" and "hourglass" are clearly similar, as are "triangle" and "non-defined triangle." However, since the heatmap provides a broader view of such proximities, this approach is arguably the most effective for quickly understanding the relationships between different limit cycle forms, particularly when it is impossible to graphically represent relationships in an 8-dimensional space.

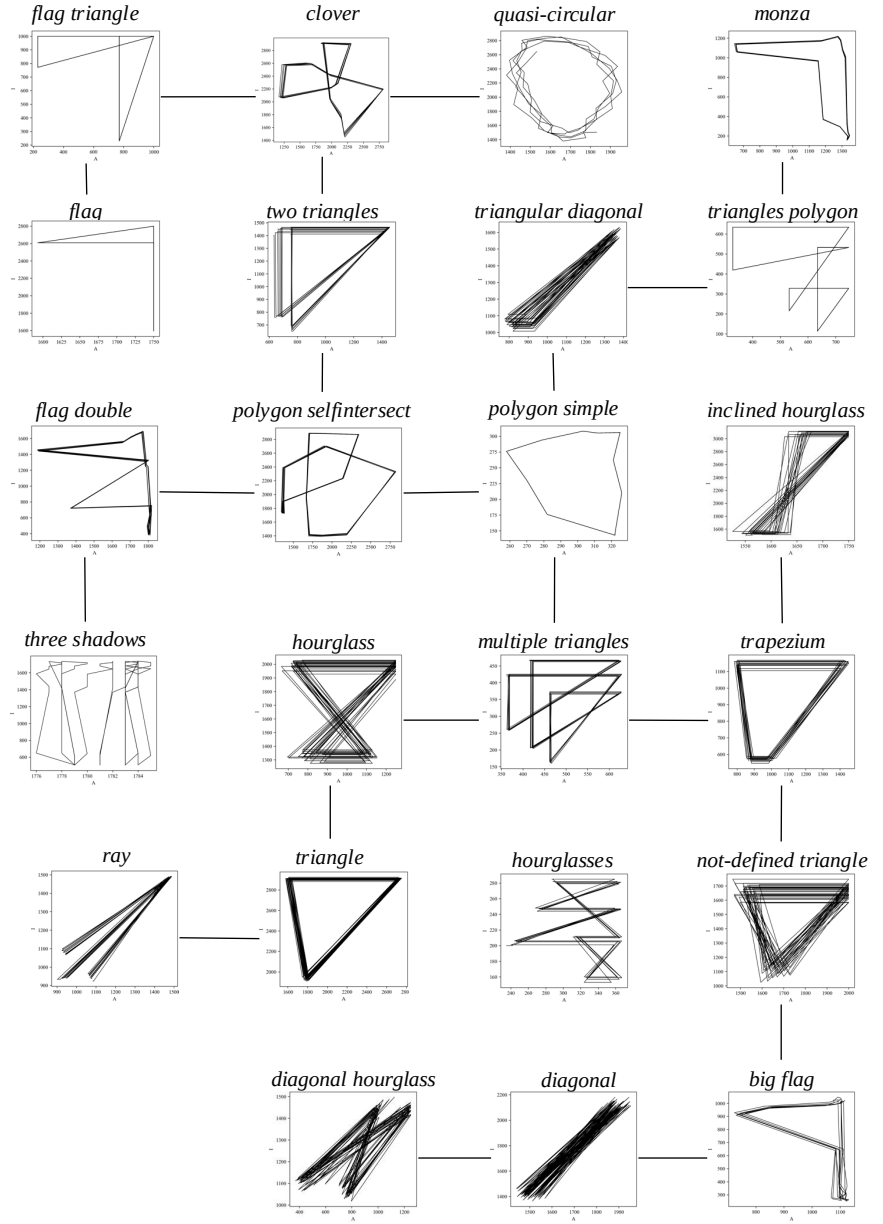


Fig 8. Cycle shapes network representation

5 Discussions

5.1 Methodological

The ability to detect limit cycles through simple numerical techniques enables the study of such phenomena even in non-analytical models such as ABMs, where traditional methods would not be applicable. Numerous ABMs exhibit emergent co-oscillatory behaviors, whose emergence often carries interesting interpretative significance [76, 77]. However, it is also likely that several models could have exhibited similar phenomena that were never investigated. As a result, the dynamic behavior of these models was explored less thoroughly, leading to a reduced generation of knowledge out of their analysis.

From a methodological perspective, it is interesting to observe the shape of the cycles that emerged in the simulation, particularly their sharp and angular form rather than smooth, rounded trajectories. One possible explanation is the relatively small number of agents involved in the simulation. Another contributing factor to the angularity of the curves is that contagion in the model is effectively event-driven, depending on the individuals attending a specific event. In this sense, the domain-specific findings of this paper may offer valuable insights for the study and management of event-driven systems.

It is nonetheless interesting to consider the possibility of first developing a taxonomy and then constructing a semantic understanding of these relationships. This perspective raises several open questions: in which other systems, if adequately studied, could such a richness of behaviors emerge? Moreover, where such behaviors exist, how much and what kind of information can be extracted through the definition of semantically meaningful proximity networks of limit cycles with different shapes? These remain open questions, and addressing them will likely require moving beyond the study of a single case — which remains valuable for testing the methodology and understanding its implications — toward applying the approach to multiple diverse cases and deductively assessing the validity and the relevance of the extracted information.

Finally, for scientists interested in the dynamics of complex systems, it is noticeable to highlight the variety of the dynamics that emerge from such a simple system. To the best of our knowledge, no comparable system exhibits such a dynamically wide range of behaviors with such a simple underlying behavior. This is of significant importance, as this detailed study may have only scratched the surface of what can be learned from this dynamic system. For instance, we have not yet defined a framework of functional relationships between the various forms — a task beyond the scope of this work but one that could further enhance our understanding of the foundations of complex systems.

Naturally, some limitations exist, the most significant being the computational cost of the methodology. Generating sufficiently long time series across a large number of simulations could require substantial time and resources. For example, in this study, only a very small percentage of simulations exhibited cyclic behavior. Although there are more targeted methodologies available to identify regions of the parameter space associated with limit cycles [16], a large number of simulations is still necessary to reliably capture these phenomena.

5.2 Case Study: Epidemiological Extension of the El Farol Bar Problem

The results table (Table 3) suggests that population size n has a positive effect on the emergence of cycles. From a policymaker's perspective, this could indicate that a potential strategy for managing epidemics similar to the one modeled in social contexts might involve compartmentalizing the social system, which means dividing it into smaller, more manageable subsystems could potentially allow the epidemic to extinguish rather than become endemic with oscillatory behavior. Similarly, lowering the threshold at which individuals perceive the bar as overcrowded can reduce the system's cyclicity, potentially leading to the extinction of the epidemic. Thus, an indirect policy measure could involve making crowded indoor spaces less appealing, effectively discouraging high attendance levels in such environments.

The same results table shows consistent signs across the various parameters, with the notable exception of the threshold for self-perception of infection. This parameter has a positive effect on the persistence of the infection but a negative effect on the presence of cyclicity. This implies that individual behavior not only influences the survival of the infection but also the likelihood of it exhibiting cyclic behavior, which would make it more predictable to some extent. This finding is significant because it suggests that individual perception plays a crucial role in managing epidemics in social contexts. Policymakers might therefore consider focusing on enhancing individual awareness and perception of infection as an essential strategy for both limiting the spread and influencing the system's dynamical properties.

Another noticeable aspect is the role of short-term memory in the decision-making process of individuals. Naturally, the more reactive individuals are to perceiving information, the easier it becomes to manage the epidemic, as this increases the likelihood of its extinction. However, with the loss of cyclicity, if the infection does not extinguish, its unpredictability increases. For a policymaker, this would make managing such a system considerably more complex.

From the perspective of a policymaker, managing such a system relies on understanding the phenomenon and identifying mechanisms that can be applied to regulate it. While this work does not address the implementation of actions, it is highly beneficial to focus on comprehension. For instance, in what sense is it easier or harder to manage a system exhibiting "polygon simple" behavior compared to "polygon self-intersect," or "diagonal" versus "diagonal hourglass"? Two key factors can be identified. The first is the number of possible states. Intuitively, the fewer possible states a system has, the easier it is for a policymaker to achieve the desired state and predict the system's next state. In this sense, "polygon self-intersect" and "polygon simple" are equally challenging to manage, as they share the same number of possible states. However, "diagonal hourglass" is more complex than "diagonal" because it has twice the number of reachable combinations in the state space (four versus two). The second factor is the comprehension of the phenomenon itself and the time required to understand it. It is assumed that a policymaker addressing an epidemic must make decisions quickly, and so it necessitates the ability to grasp the system's behavior easily. For example, it is easier to understand the "polygon simple" case and recognize that halfway through the cycle, the system will return to its initial state if left undisturbed than to interpret the "polygon self-intersect" case. This difference may relate to the Kolmogorov complexity of the processes needed to generate the specific curves without the underlying generative model. Further studies could be conducted in this direction.

Across all the results presented, a co-effect between the epidemiological and social components is consistently observed. The existence of this co-effect suggests that analyses should always consider multiple factors. For instance, calculating the R_0 parameter is important but should be complemented by some form of behavioral testing to enable appropriate interventions. Understanding the presence of this co-effect is also critical for a policymaker aiming to influence the system effectively. What actions should be taken? Should decisions focus on reducing contagion while avoiding the creation of cyclic behaviors? Understanding the existence of a two-dimensional approach is essential to balance the management of both the epidemiological and social dimensions of the system.

Regarding the domain specific results, this work has some limitations which are necessary to highlight for clearly define the applicability of the results. The first is that it is based on a simulation conducted in an in silico system without a direct equivalence in reality. This does not mean that its results cannot exist in the empirical world, but before being applied by a policymaker in a real scenario, they should be empirically validated. Secondly, in this specific case the clustering of the limit cycles was performed manually. This was done primarily to avoid introducing new complications to the proposed research pattern by adding a non-supervised machine learning classifier. Nevertheless, it is worth highlighting that future developments could include this part, especially to automate the whole process and make it quicker to implement and use. Thirdly, the precision of the cycle detection procedure was not measured, but this is not the point of the present work. The aim is not to provide statistically certified estimates how many cycles emerges in the system, but to identify whether relevant cyclic behaviors emerge and to characterize their qualitative structure. Also, given that the model is an inherently stochastic, non-analytical agent-based system, the notion of precision in the classical sense could be conceptually misplaced. From a policy and interpretive perspective, we consider the knowledge that cycles exist and how they manifest is already valuable. There are also limitations related to the model's adherence to certain modeling assumptions, such as the homogeneity of agents, particularly concerning their behavior. However, it was not deemed essential to address this aspect in the initial analysis, as it is intuitive that with a larger number of agents, behavioral heterogeneity would likely average out in the mean-field dynamics. For this reason, the specific homogeneous configuration was chosen.

6 Conclusions

This study builds on a pre-existing agent-based model [30] that extends the classic El Farol Bar problem [31] with an epidemiological dimension, where individuals also face infection risks. A subset of the simulation data was collected and analyzed to detect potential limit cycles using a custom data-driven methodology. Unlike traditional approaches, this method did not involve analytical examination of the system's formalization but instead extracted patterns directly from the simulation output.

The paper makes three main contributions: it introduces a new methodology for detecting limit cycles, providing a new perspective on identifying recurring dynamic patterns in complex systems; it proposes a framework for analyzing the relationships among different limit cycle shapes generated by the same model under different parameter configurations; it discusses the practical implications of the results, showing how the analysis of limit cycles can provide actionable information for domain experts.

The main directions for future developments of this work, beyond the empirical

studies necessary to validate the results, could include the following. First, quantifying the amount of information, in bits, required by each agent to make a decision, analyzed in terms of the system's entropy. Second, extending the study to longer-term infections involving more events, such as AIDS, where the social and epidemiological co-dynamics play a critical role. Third, exploring how a policymaker should act within this system, investigating whether specific optimal strategies exist among the available options, and assessing how the optimality of these strategies depends on the context. Finally, expanding the methodology used to identify limit cycles, testing its applicability beyond the specific context of this study to determine its validity in other scenarios.

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